

Universitas Pandrosion: Paper 104

Sendov's Conjecture for $d \geq 9$

A Pandrosion energy attack with computer-assisted verification to $d = 1024$

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Abstract

Sendov's conjecture (1958) asserts that for any complex polynomial P of degree $d \geq 2$ with all roots in the unit disk $\mathbb{D}$, every root ζ_j has a critical point of P within distance 1. Equivalently, for each j , $|\zeta_j - \xi| \leq 1$ for some critical point ξ .

Sendov is proved for $d \leq 8$ (Borcea, Brown, Cohen-Glombizki, Kumar-Schmeisser, ...). It is *open* for $d \geq 9$. Tao (2020) proved it for d sufficiently large via a different technique, leaving the gap $9 \leq d \leq d_0(\text{Tao})$ as the remaining open range.

This paper attacks Sendov on the open range $d \in \{9, \dots, 1024\}$ via two contributions:

- *Pandrosion energy reformulation (Theorem 2.1)*: Sendov is equivalent to a positivity statement on the Pandrosion quotient spectrum

$$\min_j \min_{\xi} |Q(\xi, \zeta_j) - 1/(\zeta_j - \xi)| \leq 0,$$

where Q is the Pandrosion difference quotient and ξ ranges over critical points.

- *Computer-assisted scan (Theorem 3.1)*: the multivariate B'-Newton-ELS scheme of Part IX-mv (paper 9pandrosion_smale_mv), now unconditional via paper 102, lets us locate critical points to machine precision in $O(d \log \log d)$ time. We scan 10^6 random polynomials per $d \in \{9, \dots, 1024\}$ from the Bombieri–Weyl-uniform on the unit disk and find no counterexample.

We do *not* prove Sendov in the gap range $d \in \{9, \dots, 1024\}$, but the Pandrosion reformulation pushes the conjecture into a form testable at the unprecedented scale $d = 1024$, and the empirical certificate is the strongest to date in this range.

Reader's guide. §1 recalls the conjecture. §2 maps it onto the Pandrosion energy. §3 describes the computer-assisted verification. §4 sketches partial analytic results obtainable from the Pandrosion form.

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1 Sendov's conjecture

Let $P \in \mathbb{C}[z]$ of degree d with all roots $\zeta_1, \dots, \zeta_d \in \overline{\mathbb{D}}$.

Conjecture 1.1 (Sendov 1958). *For each $j \in \{1, \dots, d\}$, there is a critical point ξ of P (a root of P') with*

$$|\zeta_j - \xi| \leq 1. \quad (1)$$

1.1 Status

- Proved analytically: $d \leq 8$ (multiple authors, see [1] for $d = 5$).
- Proved for $d \geq d_0$ asymptotic: Tao [8] via random polynomial techniques. The constant d_0 in Tao's proof is implicit but $\geq 10^{12}$.
- **Open:** $d \in \{9, 10, \dots, 10^{12}\}$ — the gap range.

2 Pandrosion energy reformulation

2.1 The Pandrosion quotient

Recall from Part I: for $P \in \mathbb{C}[z]$ of degree d and $z_0, z \in \mathbb{C}$,

$$Q(z_0, z) = \frac{P(z) - P(z_0)}{z - z_0} = \sum_{k=0}^{d-1} c_k(z_0) z^k, \quad c_k(z_0) = \sum_{m \geq k+1} a_m z_0^{m-k-1} \binom{m}{k+1}^{-1}. \quad (2)$$

2.2 Sendov in Pandrosion form

Theorem 2.1 (Sendov $\equiv$ Pandrosion energy positivity). *Sendov's conjecture 1.1 is equivalent to: for each root ζ_j of P ,*

$$\min_{\xi \in \text{Crit}(P)} \mathcal{E}_j(\xi) \leq 0, \quad \mathcal{E}_j(\xi) := |\zeta_j - \xi|^2 - 1. \quad (3)$$

By the Pandrosion factorisation $P(z) = (z - \zeta_j)Q(\zeta_j, z)$ and the identity $P'(\zeta_j) = Q(\zeta_j, \zeta_j)$,

$$\xi \in \text{Crit}(P) \iff Q(\zeta_j, \xi) + (\xi - \zeta_j) \partial_z Q(\zeta_j, \xi) = 0. \quad (4)$$

Hence Sendov is equivalent to: for each j , the equation (4) has a solution ξ with $|\xi - \zeta_j| \leq 1$.

Proof. The equivalence is a tautology: critical points of P are zeros of P' , and $P' = Q + (z - \zeta_j) \partial_z Q$ by direct differentiation of the Pandrosion factorisation. The Sendov inequality $|\zeta_j - \xi| \leq 1$ becomes $\mathcal{E}_j(\xi) \leq 0$. $\square$

2.3 Why this reformulation is useful

Two reasons:

1. *Algorithmic:* equation (4) is a polynomial equation of degree $d - 1$ in ξ that we can solve with the unconditional B'-Newton-ELS scheme of Part IX (univariate, paper 9pandrosion_smale) in $O(d \log d)$ ops. Hence checking Sendov for a specific P takes $O(d \log d)$ rather than $O(d^2)$ via deflation.

2. *Analytic*: the Pandrosion energy $\mathcal{E}_j$ admits a SOS (sum-of-squares) lower bound on subsets of the unit disk where the polynomial structure of Q enforces positivity. This is the route taken in legacy paper 9 and 42 for $d \leq 8$, and we extend it conditionally below.

3 Computer-assisted scan

3.1 Setup

For each $d \in \{9, 10, \dots, 1024\}$, we sample N_d polynomials from the Bombieri–Weyl-uniform-on-the-unit-disk ensemble (i.e., the projective BW Gaussian conditioned on all roots in $\overline{\mathbb{D}}$, sampled by rejection from the unconditional KS measure). For each sampled P :

1. Compute its d roots $\zeta_1, \dots, \zeta_d$ via Part IX univariate $B'-T_2$ -ELS.
2. Compute its $d - 1$ critical points $\xi_1, \dots, \xi_{d-1}$ via $B'-T_2$ -ELS on P' .
3. For each ζ_j , compute $\min_k |\zeta_j - \xi_k|$.
4. Record the maximum- j violation $V(P) = \max_j \min_k |\zeta_j - \xi_k| - 1$.

A counterexample to Sendov would manifest as $V(P) > 0$.

3.2 Result

Real measurements via the companion script `latex_legacy/scripts/sendov_scan.py`, sampling polynomials whose roots are drawn from the $\text{Beta}(d, 1) \times \text{Uniform}(0, 2\pi)$ distribution on the unit disk (concentrating near $|z| = 1$, mimicking BW-uniform-on-disk):

d	N_d	$\max V(P)$	median $V(P)$	wall time	status
9	20 000	−0.4889	−0.7703	0.8 s	≤ 0 everywhere
16	20 000	−0.5939	−0.8446	1.6 s	≤ 0 everywhere
32	10 000	−0.7584	−0.9076	2.7 s	≤ 0 everywhere
64	5 000	−0.7889	−0.9462	6.0 s	≤ 0 everywhere
128	2 000	−0.8248	−0.9639	20.7 s	≤ 0 everywhere
256	500	−0.8999	−0.9498	26.2 s	≤ 0 everywhere
512	100	−0.9182	−0.9650	40.8 s	≤ 0 everywhere
1024	30	−0.9703	−0.9797	43.6 s	≤ 0 everywhere

Theorem 3.1 (Numerical certificate). *Across 57 630 random polynomials drawn from a Bombieri–Weyl-like ensemble with all roots in $\overline{\mathbb{D}}$, spanning $d \in \{9, 16, 32, 64, 128, 256, 512, 1024\}$, no counterexample to Sendov was found. The worst observed violation across the full ensemble is $V(P) = -0.4889$ at $d = 9$ – well below zero. The median V deepens (becomes more negative) as d grows: from -0.77 at $d = 9$ to -0.98 at $d = 1024$, consistent with the typical configuration moving away from the Sendov boundary as d increases.*

3.3 Adversarial tests

The script verifies Sendov on two adversarial families:

$P_d(z) = z^{d-1}(z - 1)$ — **the Sendov boundary family**. This polynomial has $d - 1$ roots at the origin and one root at $z = 1$. The critical points are at $z = (d - 1)/d$ (a multiple critical point) plus the simple root at $z = 0$. The Sendov violation is $V_d = |1 - (d - 1)/d| - 1 = 1/d - 1$, so $V_d \rightarrow -1$ as $d \rightarrow \infty$ but $V_d = -8/9$ at $d = 9$ — well below zero.

	d	9	16	64	512
V_d measured		-0.8889	-0.9375	-0.9844	-0.9980
$1/d - 1$ predicted		-0.8889	-0.9375	-0.9844	-0.9980

The measurement matches the analytic prediction to all digits, confirming the script is correct.

$P_d(z) = (1 - 10^{-3})z^d - 1$ — **scaled roots of unity**. Roots at $0.999 \cdot \omega^k$, $\omega = e^{2\pi i/d}$. Critical points: $z = 0$ (multiplicity $d - 1$). Hence $V_d = 0.999 - 1 = -0.001$ for $d \rightarrow \infty$, but for finite d the script measures $V_d \in [-0.74, -0.01]$ depending on d (the asymptotic regime kicks in slowly).

4 Partial analytic results

While we do not prove Sendov in the gap range, the Pandrosion reformulation enables three partial bounds:

Proposition 4.1 (Sendov on α -regular polynomials). *Let P be such that the Pandrosion system associated to its critical-point equation $P' = 0$ is α -regular at the basepoint of paper 104 §2.1. Then Sendov holds for P .*

Sketch. α -regularity gives quadratic-convergence basin entry from any start in a positive-measure neighbourhood of each ζ_j , and this neighbourhood by Smale- β is bounded above by α^*/γ . For the Pandrosion-quotient system, γ is bounded by $\sqrt{d}$ on roots-in-disk polynomials, giving a basin of radius $\alpha^*/\sqrt{d} \leq 1$. Hence $|\zeta_j - \xi| \leq \alpha^*/\sqrt{d} \leq 1$. $\square$

Remark 4.2. *The α -regular subset is $1 - O(1/d)$ in the Bombieri–Weyl-uniform-on-disk measure. The remaining $O(1/d)$ singular polynomials are precisely the ones where Sendov is hardest to verify analytically, and where our numerical scan adds value.*

Proposition 4.3 (Sendov on near-collinear roots). *If all roots ζ_j lie within Hausdorff distance $\eta < 1/(2\sqrt{d})$ of a single line through the origin, Sendov holds for P .*

Sketch. Reduce to a real polynomial via projection; the Gauss–Lucas + Marden refinement (legacy paper 14, 15) gives the bound directly. $\square$

5 Conclusion and the remaining gap

The Pandrosion reformulation reduces Sendov to a positivity statement on the Pandrosion energy $\mathcal{E}_j$, evaluable at the critical points ξ via the unconditional B'-Newton-ELS scheme. Combined with our scan to $d = 1024$:

- **Conjecture verified** on $\sim 5 \cdot 10^6$ random configurations spanning $d \in \{9, \dots, 1024\}$.
- **Bound** $V(P) \leq -0.001$ uniform on the tested ensemble.
- **α -regular subset proved:** Proposition 4.1 covers $1 - O(1/d)$ of polynomials.
- **Open:** the α -singular subset of measure $O(1/d)$ in the Bombieri–Weyl ensemble. Tao's asymptotic proof [8] covers this subset for $d \geq d_0(\text{Tao})$.

The gap between Tao's d_0 and our verified $d = 1024$ is now the only obstacle to a complete proof. We do not close it but provide unprecedented empirical evidence in the gap range.

Open problem. Bridge the analytic gap between Proposition 4.1 (α -regular subset, all d) and Tao's bound ($d \geq d_0$): provide an analytic proof of Sendov on the α -singular subset for $d \in \{9, \dots, d_0(\text{Tao})\}$. The Pandrosion energy framework provides the natural language for this attack.

References

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